



Assignment 09: Working with Functions in Python



Instructions

1. Write each program in a separate Python file.
 2. Use meaningful function names.
 3. Call the function after defining it.
 4. Display clear output messages.
 5. Use parameters and return values wherever required.
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1 Create Your First Function

Problem:

Create a function named `welcome()` that displays the message:

```
Welcome to Python Functions
```

Call the function to display the message.

Example Output

```
Welcome to Python Functions
```



2 Function to Add Two Numbers

Problem:

Create a function `add_numbers(a, b)` that takes two numbers as parameters and displays their sum.

Example Output

```
First Number = 10  
Second Number = 20  
  
Sum = 30
```



3 Function to Calculate Area of a Rectangle

Problem:

Create a function `area(length, width)` that calculates and displays the area of a rectangle.

Formula

```
Area = Length × Width
```

Example Output

```
Length = 10  
Width = 5  
  
Area = 50
```



Function to Check Even or Odd

Problem:

Create a function `check_even_odd(number)` that checks whether a number is even or odd.

Example Output

```
Number = 15  
  
Odd Number
```



Function to Find the Largest of Two Numbers

Problem:

Create a function `largest(a, b)` that finds and displays the larger number.

Example Output

```
First Number = 25  
Second Number = 40  
  
Largest Number = 40
```



Function to Calculate Square of a Number

Problem:

Create a function `square(number)` that returns the square of a given number.

Example Output

```
Number = 8  
  
Square = 64
```

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Function to Display a Multiplication Table

Problem:

Create a function `table(number)` that displays the multiplication table of a number from 1 to 10.

Example Output

```
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
...  
5 x 10 = 50
```

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Function to Calculate Total and Average Marks

Problem:

Create a function `calculate_result(marks)` that accepts a list of marks and displays:

- Total Marks
- Average Marks

Example Output

```
Marks = [78, 85, 92, 67, 88]  
  
Total = 410  
Average = 82.0
```



Bonus Question

Student Grade Calculator

Problem:

Create a function `grade(marks)` that returns the grade according to the following rules:

Marks	Grade
80 and above	A
60 to 79	B
40 to 59	C
Below 40	D

Display the returned grade.

Example Output

```
Marks = 72
```

```
Grade = B
```

Learning Outcomes

After completing this assignment, students will be able to:

✓ Define and call functions ✓ Pass arguments to functions ✓ Return values from functions ✓ Use functions with loops and conditions ✓ Organize programs into reusable blocks of code ✓ Build modular Python programs using functions